WORKED SOLUTION

STEP II Specimen paper, Question 15

Probability and Statistics • Expectation and Sample Means

STEP II Specimen paper, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q15

The key idea

The king fines a huge number of vendors every day, so by the law of large numbers his income per vendor per day is, to all intents, the *expected* fine. We therefore need the expected fine under each scheme. The first two schemes are about the mean of a sample of normal variables, whose distribution is again normal with a smaller variance. The third is a waiting-time problem, so it calls for the geometric distribution.

Solution

Let $W_1, W_2, \ldots$ be the weights (in grams) of the kippers in the selected carton. The carton holds an essentially random selection of kippers, so the W_i are independent, each $W_i \sim \mathrm{N}(200, 10^2)$. Choosing kippers from the carton at random doesn't depend on their weights, so any kippers we pick out, in any order, are still independent $\mathrm{N}(200, 10^2)$ variables.

We use the standard fact that if $W_1, \ldots, W_m$ are independent $\mathrm{N}(\mu, \sigma^2)$, then their mean satisfies

$$\overline{W} = \frac{1}{m}\sum_{i=1}^{m} W_i \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{m}\right).$$

Part (i)

Here $m = 625$, so $\overline{W} \sim \mathrm{N}\left(200, \frac{100}{625}\right)$, and the standard deviation is $\frac{10}{25} = 0.4$. Then

$$\mathrm{P}(\overline{W} < 199) = \Phi\left(\frac{199 - 200}{0.4}\right) = \Phi(-2.5) = 1 - \Phi(2.5) \approx 1 - 0.99379 = 0.00621.$$

The expected fine is

$$1500 \times 0.00621 \approx 9.3 \text{ crowns.}$$

Part (ii)

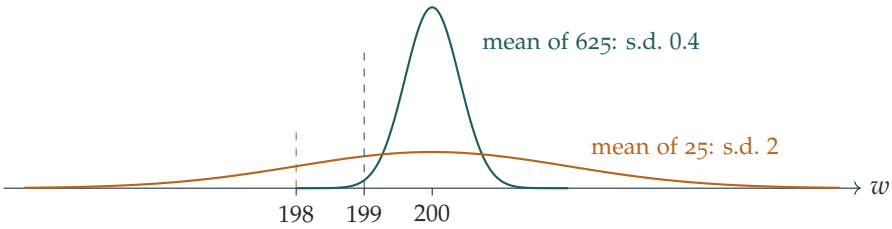
Now $m = 25$, so $\overline{W} \sim \mathrm{N}\left(200, \frac{100}{25}\right)$, with standard deviation 2. Then

$$\mathrm{P}(\overline{W} < 198) = \Phi\left(\frac{198 - 200}{2}\right) = \Phi(-1) = 1 - \Phi(1) \approx 1 - 0.84134 = 0.15866,$$

and the expected fine is

$$100 \times 0.15866 \approx 15.9 \text{ crowns.}$$

The picture shows why the smaller fine wins. A threshold one gram below the mean is 2.5 standard deviations away for the average of 625 kippers, but a threshold two grams below is only one standard deviation away for the average of 25.



Part (iii)

Let N be the number of kippers removed. Each kipper weighs more than 200 g with probability $\frac{1}{2}$, since 200 is the mean of a symmetric distribution and $\mathrm{P}(W = 200) = 0$. The kippers drawn are independent, so N is the number of trials up to and including the first success in independent trials with success probability $\frac{1}{2}$. In other words, N is geometric:

$$\mathrm{P}(N = n) = \left(\tfrac{1}{2}\right)^n, \qquad n = 1, 2, 3, \ldots$$

(Strictly, the carton runs out after 625 kippers, but the chance that none of them exceeds 200 g is 2^{-625}, which is utterly negligible.)

Ask yourself How do we find $\mathrm{E}[N(N-1)]$ without first computing $\mathrm{E}[N^2]$?

Differentiate the geometric series twice. For $|x| < 1$,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \Longrightarrow \quad \sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}.$$

Multiplying by x^2 and putting $x = \frac{1}{2}$ (the $n = 1$ term is zero anyway):

$$\mathrm{E}[N(N-1)] = \sum_{n=1}^{\infty} n(n-1)\left(\tfrac{1}{2}\right)^n = \left(\tfrac{1}{2}\right)^2 \cdot \frac{2}{\left(\frac{1}{2}\right)^3} = 4.$$

(Equivalently, $\mathrm{E}[N] = 2$ and $\mathrm{Var}(N) = \frac{1-p}{p^2} = 2$ with $p = \frac{1}{2}$, so $\mathrm{E}[N^2] = 6$ and $\mathrm{E}[N^2] - \mathrm{E}[N] = 4$.) So the expected fine is

$$\mathrm{E}[3N(N-1)] = 3 \times 4 = 12 \text{ crowns.}$$

Conclusion

The expected fines per vendor per day are about 9.3, 15.9 and 12 crowns for schemes (i), (ii) and (iii). Because vast numbers of cartons are inspected, the king's total income is very close to the number of inspections times the expected fine. He should therefore choose **scheme (ii)**. It raises roughly 70% more than scheme (i) and about a third more than scheme (iii).

Checks

- Scheme (iii) can't be checked against a table, but the series is easy to add up: $\frac{3\cdot2}{4} + \frac{3\cdot6}{8} + \frac{3\cdot12}{16} + \cdots = 1.5 + 2.25 + 2.25 + 1.875 + \cdots$. The partial sums climb quickly towards 12.
- A simulation of 10^5 cartons of 625 normal kippers, run through all three schemes, gave average fines of about 10.2 (from 2×10^4 cartons, so noisy), 15.9 and 11.9 crowns. These are close to 9.31, 15.87 and 12.

Common slips

- Using a standard deviation of $\frac{10}{625}$ or $\frac{100}{25}$ for the sample mean. The *variance* is divided by m, so the standard deviation is divided by $\sqrt{m}$.
- Treating N as starting from 0 (counting failures only). Here n includes the heavy kipper, so $\mathrm{P}(N = 1) = \frac{1}{2}$ and the fine is then zero.
- Comparing probabilities of being fined instead of expected fines: scheme (iii) fines a vendor (with $n \geq 2$) half the time, but usually only 6 crowns.

Beyond STEP

The justification "vast quantities, so use the expectation" is the law of large numbers, and the fact that $\overline{W}$ is normal with variance σ^2/m comes with the central limit theorem, which makes it approximately true even when the individual weights aren't normal. A cautious king might also weigh the spread: scheme (i), a rare large fine, has by far the largest variance. The idea of finding factorial moments like $\mathrm{E}[N(N-1)]$ by differentiating a series leads to probability generating functions, a central tool in first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers all of this clearly.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.